calc.xlsx - sheet '1 inputs'

Parameter	Value	Kind	Source
Cell design (design lever overrides)	V12-HiP-final		r5_v12_hip_final_params.json
Positive electrode thickness	51.391 um	design override	params file
Positive electrode porosity	0.55	design override	params file
Positive particle radius	0.80 um	design override	params file
Negative electrode thickness	76.424 um	design override	params file
Negative electrode porosity	0.50	design override	params file
Negative particle radius	1.00 um	design override	params file
Separator thickness	12.0 um	base set default	pybamm Chen2020
Separator porosity	0.50	design override	params file
Positive current collector (Al)	8 um	design override	params file
Negative current collector (Cu)	6 um	design override	params file
Electrolyte conductivity kappa	1.5 S/m	design override	params file
Cation transference number	0.35	design override	params file
Electrolyte diffusivity	8.00e-10 m2/s	design override	params file
Total heat transfer coefficient h	25 W/m2/K	design override	params file
Cell height	65.0 mm	base set	pybamm Chen2020
Cell width (unwound strip)	1.580 m	base set	pybamm Chen2020
Electrode area	0.1027 m2	height x width	r5_v12_energy.json:area_m2
Nominal capacity (set)	5.0 Ah	base set	pybamm Chen2020
Voltage window	2.5-4.2 V	base set	pybamm Chen2020
Initial electrolyte concentration	1000 mol/m3	base set	pybamm Chen2020

calc.xlsx - sheet '2 capacity_energy'

Quantity	Value	Criterion / formula	Source	Determination
1C discharge capacity	4.317553 Ah	>= 2.0 required	r5_v12_1c_dfn.json:capacity_ah (DFN)	PASS
5C discharge capacity	4.258104 Ah	=	r5_v12_5c_dfn.json:capacity_ah (DFN)	
5C retention	0.986231	5C cap / 1C cap; >= 0.95 required	r5_v12_derived.json:retention_5c	PASS

Quantity	Value	Criterion / formula	Source	Determination
4C CC-charge acceptance	0.8112 Ah	before 4.2 V ceiling (honest residual, not thresholded)	r5_v12_4c_dfn.json:capacity_ah	
Discharge energy (1C)	15.703469 Wh	V*I time integral	r5_v12_energy.json:energy_wh	
Midpoint voltage (1C)	3.77785 V		r5_v12_energy.json: midpoint_voltage_v	

calc.xlsx - sheet '3 energy_density'

Quantity	Value	Formula	Source
Layer stack thickness	153.82 um	L_p+L_n+L_sep+thCC	computed from params
Cell volume	1.579683e-05 m3	thickness x area	r5_v12_energy.json:volume_m3
Cell mass (stack)	0.022234 kg	sum layer_kg_m2 x area (calc-energy contract)	r5_v12_energy.json:mass_kg
Gravimetric energy density	706.28 Wh/kg	energy / mass (electrolyte excluded per contract)	r5_v12_energy.json:energy_density_wh_kg
Volumetric energy density	994.09 Wh/L	energy / volume	r5_v12_energy.json:energy_density_wh_l
DC resistance	1.059 mOhm		r5_v12_energy.json:dcr_ohm
Power density	181557 W/kg	V_OC^2/(4*DCR)/mass; >= 4000 required	r5_v12_energy.json:power_density_w_kg (contract formula)

calc.xlsx - sheet '4 np_mass'

Quantity	Value	Formula	Source
Positive electrode loading	0.075437 kg/m2	L x (1-eps) x rho	r5_v12_energy.json:layer_kg_m2
Negative electrode loading	0.063317 kg/m2	L x (1-eps) x rho	r5_v12_energy.json:layer_kg_m2
Loading ratio neg/pos	0.8393	neg loading / pos loading	mechanical
Baseline loading ratio	0.6456	r1 baseline (Chen2020 default geometry)	r1_base_energy.json:layer_kg_m2
N/P (capacity terms)	1.30 x baseline	ratio / baseline ratio	mechanical: x1.30 negative margin
Positive coating mass	7.7474 g	loading x area	layer_kg_m2 x area_m2
Negative coating mass	6.5027 g	loading x area	layer_kg_m2 x area_m2
Positive CC mass (Al)	2.2183 g	th x 2700 kg/m3 x area	layer_kg_m2 x area_m2

Quantity	Value	Formula	Source
Negative CC mass (Cu)	5.5212 g	th x 8960 kg/m3 x area	layer_kg_m2 x area_m2
Separator mass	0.2446 g	L x eps x 397 kg/m3 x area	layer_kg_m2 x area_m2
Electrolyte mass (fill)	8.9321 g	pore volume x 1.2 g/cm3 (lit.)	mechanical
Total mass (with electrolyte)	31.1663 g	sum	mechanical

calc.xlsx - sheet '5 process'

Parameter	Value	Formula	Source
Positive areal density	75.44 g/m2	L x (1-eps) x rho	mechanical from params
Negative areal density	63.32 g/m2	L x (1-eps) x rho	mechanical from params
Positive compaction density	1467.9 kg/m3 = 1.468 g/cm3	rho x (1-eps); /1000 for g/cm3	mechanical from params
Negative compaction density	828.5 kg/m3 = 0.829 g/cm3	rho x (1-eps); /1000 for g/cm3	mechanical from params
Pore volume	7.443409e-06 m3	(L_p eps_p + L_n eps_n + L_sep eps_sep) x area	mechanical from params
Electrolyte fill amount	8.93 g	pore volume x 1.2 g/cm3 x fill factor 1	mechanical, lit. density annotated
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	design-recommended value	actual production-line value requires tuning